

Intro to MD: Setting Up a System in GROMACS

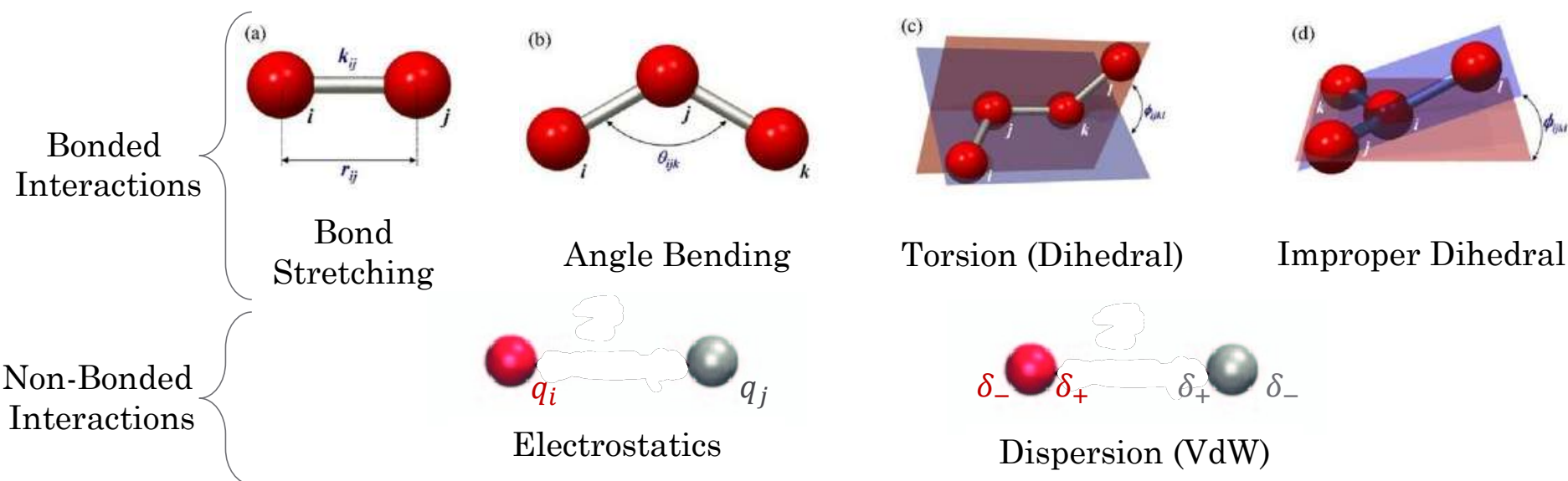
Feb 11th 2025 Group Meeting

Outline

- What is Molecular Dynamics?
- What Do You Need To Set Up A GROMACS Simulation?
 - Forcefield/Topology File
 - Coordinate File
- Tutorial Example

What is Molecular Dynamics?

It's Just Wiggles in Time



For an atom ' i ':

$$E_i = E_{\text{Bonded},i} + E_{\text{Unbonded},i}$$

$$\mathbf{F}_i = m_i \mathbf{a}_i = - \frac{\partial E_i}{\partial \mathbf{x}}$$

$$\mathbf{v}_i = \frac{d\mathbf{a}_i}{dt}; \quad \mathbf{x}_i = \frac{d\mathbf{v}_i}{dt}$$

What you need to start

A forcefield ($\mathbf{F}_i(\mathbf{x})$)
Initial coordinates ($\mathbf{x}_0$)

What Do You Need To Set Up A GROMACS Simulation?

A Forcefield – Topology File `.top`

```
[ defaults ]
; nbfunc      comb-rule      gen-pairs      fudgeLJ fudgeQQ
1             2              yes             0.5       0.83333333

[ atomtypes ]
; name      at.num      mass      charge ptype      sigma      epsilon
c3          6      12.010000  0.00000000 A      0.33996695      0.4577296
na          7      14.010000  0.00000000 A      0.32499985      0.71128
cc          6      12.010000  0.00000000 A      0.33996695      0.359824

[ moleculetype ]
; Name      nrexcl
C2M          3

[ atoms ]
; nr      type      resnr residue atom      cgnr      charge      mass      typeB
; residue 1 C2M rtp C2M q 1.0
1         c3         1      C2M      C1         1 -0.12610000 12.010000 ; qt
2         c3         1      C2M      C2         2 0.01830000 12.010000 ; qtc

[ bonds ]
; ai      aj funct      c0      c1      c2      c3
6         7      1      0.13802 356309.440000
6         8      1      0.14629 274219.360000
5         6      1      0.13802 356309.440000
```

```
[ pairs ]
; ai      aj funct      c0      c1      c2      c3
4         8      1
3         8      1

[ angles ]
; ai      aj      ak funct      c0      c1      c2      c3
7         6      8      1      126.4600543 517.979200
5         6      7      1      128.0100547 531.368000
-         -         -         -         -         -         -         -

[ dihedrals ]
; ai      aj      ak      al funct      c0      c1      c2
5         4         3         7      1      180.0000771 7.1128000 2
4         3         7         6      1      180.0000771 7.1128000 2
4         5         6         7      1      180.0000771 7.1128000 2
4         5         6         8      1      180.0000771 7.1128000 2

2         3         7      16      1      180.0000771 7.1128000 2
16        3         7         6      4      180.0000771 4.6024000 2
4         15        5         6      4      180.0000771 4.6024000 2

[ system ]
; Name
Generic title

[ molecules ]
; Compound      #mols
C2M              250
TF2              250
```

What Do You Need To Set Up A GROMACS Simulation?

Initial Coordinates – `.gro` File

GROningen MAchine for Chemical Simulation

8500

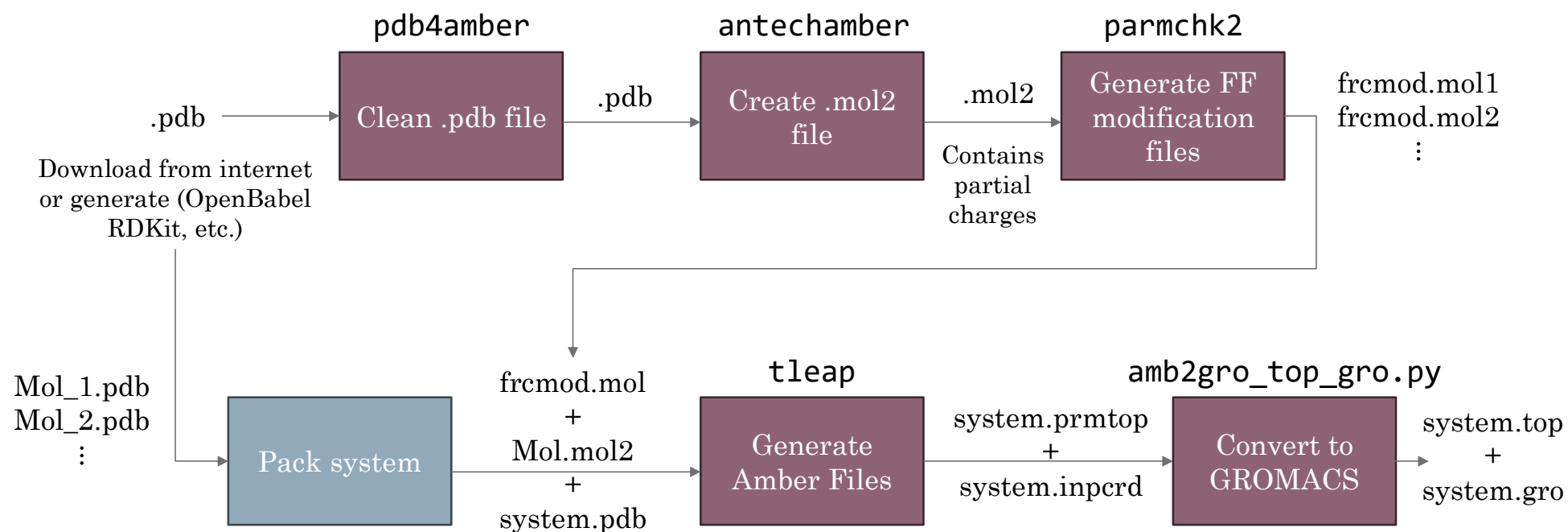
1C2M	C1	1	3.891	6.024	6.184
1C2M	C2	2	3.853	6.097	6.312
1C2M	N1	3	3.838	6.004	6.423
1C2M	C3	4	3.720	5.943	6.463
1C2M	C4	5	3.751	5.864	6.571
1C2M	N2	6	3.886	5.879	6.594
1C2M	C5	7	3.938	5.964	6.504
1C2M	C6	8	3.963	5.813	6.699
1C2M	H1	9	3.986	5.970	6.197
1C2M	H2	10	3.903	6.094	6.102
1C2M	H3	11	3.815	5.951	6.157
1C2M	H4	12	3.931	6.170	6.338
1C2M	H5	13	3.759	6.151	6.298
1C2M	H6	14	3.629	5.964	6.410
1C2M	H7	15	3.693	5.799	6.635
1C2M	H8	16	4.042	5.995	6.497
1C2M	H9	17	4.068	5.842	6.693
1C2M	H10	18	3.922	5.843	6.796
1C2M	H11	19	3.954	5.705	6.687
2C2M	C1	20	1.523	0.572	1.857
2C2M	C2	21	1.528	0.509	1.995
2C2M	N1	22	1.646	0.424	2.010
2C2M	C3	23	1.652	0.287	1.985

250C2M	H10	4749	2.390	3.162	6.073
250C2M	H11	4750	2.281	3.183	6.213
251TF2	C1	4751	1.725	2.104	6.739
251TF2	F1	4752	1.620	2.133	6.821
251TF2	F2	4753	1.697	1.986	6.676
251TF2	F3	4754	1.830	2.077	6.823
251TF2	S1	4755	1.768	2.238	6.620
251TF2	O1	4756	1.772	2.363	6.696
251TF2	O2	4757	1.900	2.200	6.567
251TF2	N1	4758	1.655	2.227	6.509
251TF2	S2	4759	1.507	2.231	6.563
251TF2	O3	4760	1.472	2.337	6.658
251TF2	O4	4761	1.451	2.100	6.596
251TF2	C2	4762	1.427	2.279	6.406
251TF2	F4	4763	1.291	2.290	6.416
251TF2	F5	4764	1.451	2.189	6.306
251TF2	F6	4765	1.472	2.399	6.359
252TF2	C1	4766	3.046	0.133	4.805
252TF2	F1	4767	3.054	0.199	4.687
252TF2	F2	4768	3.058	-0.001	4.779
252TF2	F3	4769	2.917	0.150	4.848

500TF2	F5	8499	4.768	2.830	5.696
500TF2	F6	8500	4.741	3.044	5.663
6.90840	6.91340	6.90820			

Tutorial on Building A System

Emim-TF2 Using GAFF Parameters



Legend:

AmberTools

PackMol